

04_Basic_IP_Addressing_Subnetting – Assign Multiple Subnets and Test Connectivity

Packet Tracer – Basic IP Addressing and Subnetting

Objectives

Part 1: Build a Network with Multiple Subnets

Part 2: Configure Router Interfaces for Each Subnet

Part 3: Assign Static IP Addresses to PCs

Part 4: Test Inter-Subnet Connectivity Using Ping

Background / Scenario

In small networks, devices in different **subnets** cannot communicate directly. A **router** is required to route traffic between these networks.

In this lab, you will:

- Build a network with **two different subnets**
- Configure **router interfaces for each subnet**
- Assign **static IPv4 addresses to PCs**
- Understand **subnet masks and network separation**
- Test **communication between subnets**

This lab reinforces understanding of:

- IPv4 addressing
 - Subnet masks
 - Router interface configuration
 - Inter-subnet routing
 - Connectivity testing using **ICMP (ping)**
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Instructions

Part 1: Build the Network Topology

Step 1: Add Devices

a. Open **Cisco Packet Tracer**

b. Add the following devices:

From **End Devices**

- Drag **PC0**
- Drag **PC1**

From **Network Devices → Routers**

- Drag **Router0**

Rename the PCs:

- **PC0 → PC-A**
- **PC1 → PC-B**

Step 2: Connect the Devices

Click **Connections (lightning bolt icon)**.

Use **Copper Cross-Over cable** because PCs are connected directly to the router.

Make the following connections:

Device Port		Connected To Port
PC-A	FastEthernet0	Router0 GigabitEthernet0/0
PC-B	FastEthernet0	Router0 GigabitEthernet0/1

Wait until the **link lights turn green**.

Your physical topology is now complete.

Part 2: Configure Router Interfaces

Step 1: Configure Interface for Subnet 1

Click **Router0 → CLI**

Enter the following commands:

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/0
```

```
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

Explanation:

This interface connects to **PC-A's network (192.168.1.0/24)**.

Step 2: Configure Interface for Subnet 2

Continue configuring the second interface:

```
interface gigabitEthernet 0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
end
```

Explanation:

This interface connects to **PC-B's network (192.168.2.0/24)**.

Part 3: Configure Static IP Addressing

Step 1: Configure PC-A

Click **PC-A → Desktop → IP Configuration**

Enter the following:

Setting	Value
IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

Step 2: Configure PC-B

Click **PC-B → Desktop → IP Configuration**

Enter:

Setting	Value
IP Address	192.168.2.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1

Part 4: Verify Connectivity

Step 1: Ping Router from PC-A

Open **PC-A** → **Desktop** → **Command Prompt**

Type:

```
ping 192.168.1.1
```

Expected Result:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Step 2: Ping PC-B from PC-A

Type:

```
ping 192.168.2.10
```

Expected Result:

You should receive **4 successful replies**.

Step 3: Ping PC-A from PC-B

Open **PC-B** → **Command Prompt**

Type:

```
ping 192.168.1.10
```

Expected Result:

Reply from 192.168.1.10

This confirms **routing between subnets is working**.

Network Summary

Device	Interface	IP Address	Subnet
Router0	G0/0	192.168.1.1	192.168.1.0/24
Router0	G0/1	192.168.2.1	192.168.2.0/24
PC-A	Fa0	192.168.1.10	192.168.1.0/24
PC-B	Fa0	192.168.2.10	192.168.2.0/24

Troubleshooting Tips

✓ Verify **correct ports are used**

- PC → **FastEthernet0**
- Router → **GigabitEthernet0/0 or GigabitEthernet0/1**

✓ Ensure router interfaces are **enabled**

no shutdown

✓ Confirm **default gateway is correct**

✓ Check that **subnet masks match the network**

✓ Wait a few seconds after enabling interfaces for links to turn **green**

What You Learned

- How subnetting separates networks
- How routers connect multiple subnets
- How to configure router interfaces
- How default gateways allow inter-network communication
- How to test network connectivity using **ping**

Challenge Extensions (Optional)

- 1** Add **two more PCs** and place them in each subnet.

Example:

Subnet 1

- PC-C → 192.168.1.20

Subnet 2

- PC-D → 192.168.2.20

Test connectivity across all devices.

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- 2** Change the subnet mask to **/26 (255.255.255.192)** and redesign the addressing scheme.

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- 3** Add a **switch between PCs and router** to simulate a more realistic LAN.

✓ You have now completed a **Basic IP Addressing and Subnetting Lab**, an important foundation for **CCNA networking concepts and routing configuration**.